



Module Manual

Master of Science (M.Sc.)

Ship and Offshore Technology

Joint Master

Cohort: Winter Term 2025

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Program description

Content

The joint Master Course „Ship and Offshore Technology“ is held in cooperation with the University of Strathclyde, Glasgow, UK. The course is designed for graduate engineers or those from related disciplines who wish to acquire advanced knowledge on ship and offshore technologies. The graduates possess a critical awareness against new knowledge in their discipline, on which basis they are enabled to act responsibly in their professional and societal environment. As a result of the cooperation, graduates are specialised in offshore technology with dedicated knowledge on ships.

Career prospects

The Master course strengthens the engineering, mathematical and natural science knowledge gained during the Bachelor education and conveys competences to solve problems in a systematic, scientific and independent fashion relevant for industry and research activities. The contents concern analysis, design and implementation methods for Ships and Offshore systems. The individual selection of the elective modules allows for a certain specialization while the mandatory courses secure a solid understanding of the general basics and in the related fields. Thereby the students are able to adjust their study contents individually according to their personal preferences. Further, the solid knowledge of the general basics and knowledge in the field related to the chosen specialisation allow for a broad professional expertise and thus a wide professional applicability. The graduates can take on scientific tasks at universities or research institutes with the aim of a doctoral dissertation or find their way directly into the industry. As for the latter, they may specialize in dedicated areas and with further experience and qualification they can take over leading roles.

Learning target

The graduates can analyse problems scientifically and solve them, even though they are not typical or only partially defined with conflicting objectives; complex tasks can be solved by abstracting from on-going research and development activities in their discipline; innovative and new methods can be used to find fundamental solutions; knowledge gaps can be identified and solutions can be proposed to overcome these gaps; theoretical and experimental investigations can be planned and executed; results can be analysed critically and conclusions can be drawn; emerging technologies can be analysed and reviewed. By doing so, they can classify knowledge from different disciplines systematically and thereby cope with complex problems. Further, they are able to reflect on the non-technical aspects of their engineering tasks responsibly. They can expand on the knowledge gained and develop further competences, also with the aim to succeed with a doctoral thesis. Consequently, the key skills from the preceding Bachelor education relevant for practical engineering tasks will be expanded in this Master course.

Program structure

This master course consists of the following mandatory courses held at the University of Strathclyde:

- Offshore Engineering Practice
- Risers and Mooring Lines
- Marine Pipelines
- Dynamics of Floating Offshore Installations
- Maritime safety and risk
- Design and construction of FPSOs
- Theory and Practice of Marine CFD
- Inspection and survey
- Group Project
- Research Project

At the Hamburg University of Technology the courses are modularized and follow the university-wide standard course structure with course modules of six credit points. The Master course combines the disciplines relevant for Naval Architecture and Ocean Engineering on the basis of the preceding Bachelor studies. Essential modules are mandatory for all students to allow for an even skill level among graduates. Further, students are able to personalize their studies due to several module options. The following modules comprise the mandatory core qualification with six credit points each:

- Ship Design
- Structural Analysis of Ships and Offshore Structures
- Ship Vibration

The students further specialize by individually selecting two modules from the following options:

- Seakeeping of Ships and Laboratory on Naval Architecture
- Fatigue Strength of Ships and Offshore Structures
- Arctic Technology
- Innovative CFD Approaches
- Manoeuvrability and Shallow Water Ship Hydrodynamics
- Nonlinear Structural Analysis

Finally, the master thesis must be completed valued at 30 credits:

- Master Thesis

Core Qualification

This programme is designed for graduate engineers in naval architecture, offshore engineering, mechanical engineering and other related disciplines who wish to acquire advanced knowledge in a broad range of subjects of ship and offshore technologies.

This two year course is offered jointly with Hamburg University of Technology (TUHH) in Germany. Year 1 is completed at Strathclyde and Year 2 in Hamburg. The award is made in the name of both universities.

The first year will be held at the University of Strathclyde will provide skills in Offshore Engineering Practice, Risers & Mooring Lines, Marine Pipelines, Dynamics of Floating Offshore Installations, Maritime Safety & Risk, Design & Construction of Floating, Production, Storage & Offloading Vessels, Theory & Practice of Marine Computational Fluid Dynamics (CFD), Inspection & Survey, also while you carry out a group and research Project. The second year will be held at the Hamburg University of Technology where you gain skills in Structural Analysis of Ships & Offshore Structures, Ship Design Seakeeping of Ships & Laboratory on Naval Architecture, as well as in two of the following subjects: Ship Vibration, Non-Linear Structural Analysis, Fatigue Strength of Ships & Offshore Structures, Arctic Technology, Innovative CFD Approaches, Manoeuvrability & Shallow Water Ship Hydrodynamics. In addition you will carry out your master thesis in a specific topic at the Hamburg University of Technology.

Module M0601: Structural Analysis of Ships and Offshore Structures

Courses

Title	Typ	Hrs/wk	CP
Structural Analysis of Ships and Offshore Structures (L0272)	Lecture	2	3
Structural Analysis of Ships and Offshore Structures (L0273)	Recitation Section (small)	2	3
Module Responsible	Prof. Alexander Düster		
Admission Requirements	None		
Recommended Previous Knowledge	Mathematics I, II, III, Mechanics I, II, III, IV Differential Equations 2 (Partial Differential Equations)		
Educational Objectives	After taking part successfully, students have reached the following learning results		
Professional Competence			
<i>Knowledge</i>	Students are able to + give an overview of the basics of structural mechanics for the analysis of ships and offshore structures. + explain structural models for thin-walled structures. + specify problems of linear structural analysis, to identify them in a given situation and to explain their mathematical and mechanical background. + classify finite elements with respect to their suitability for the structural analysis of ships and offshore structures.		
<i>Skills</i>	Students are able to + model linear structural problems of ships and offshore structures. + select a suitable finite element formulation for a given linear problem of structural mechanics . + apply finite element procedures to the linear structural analysis of ships and offshore structures. + verify and critically judge the results of linear finite element computations. + transfer their knowledge of linear structural analysis with finite elements to new problems.		
Personal Competence			
<i>Social Competence</i>	Students are able to + solve problems in heterogeneous groups and to document the corresponding results. + share new knowledge with group members.		
<i>Autonomy</i>	Students are able to + assess their knowledge by means of exercises and E-Learning.		
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56		
Credit points	6		
Course achievement	None		
Examination	Written exam		
Examination duration and scale	2h		
Assignment for the Following Curricula	Naval Architecture and Ocean Engineering: Core Qualification: Compulsory Naval Architecture and Ocean Engineering: Core Qualification: Compulsory Ship and Offshore Technology: Core Qualification: Compulsory		

Course L0272: Structural Analysis of Ships and Offshore Structures	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Alexander Düster
Language	EN
Cycle	WiSe
Content	1. Introduction 2. Basic equations of elastostatics 3. Approximation procedures 4. The finite element method 5. Mechanical models and finite elements for thin-walled structures 6. Application to ships and offshore structures
Literature	[1] Alexander Düster, Structural Analysis of Ships and Offshore Structures, Lecture Notes, Technische Universität Hamburg-Harburg, 125 pages, 2014. [2] G. Clauss, E. Lehmann, C. Östergaard, M.J. Shields, Offshore Structures: Volume II, Strength and Safety for Structural Design, Springer, 1993. [3] G. Clauss, E. Lehmann, C. Östergaard, Meerestechnische Konstruktionen, Springer, 1988.

Course L0273: Structural Analysis of Ships and Offshore Structures	
Typ	Recitation Section (small)
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Alexander Düster
Language	EN
Cycle	WiSe
Content	1. Introduction 2. Basic equations of elastostatics 3. Approximation procedures 4. The finite element method 5. Mechanical models and finite elements for thin-walled structures 6. Application to ships and offshore structures
Literature	[1] Alexander Düster, Structural Analysis of Ships and Offshore Structures, Lecture Notes, Technische Universität Hamburg-Harburg, 125 pages, 2014. [2] G. Clauss, E. Lehmann, C. Östergaard, M.J. Shields, Offshore Structures: Volume II, Strength and Safety for Structural Design, Springer, 1993. [3] G. Clauss, E. Lehmann, C. Östergaard, Meerestechnische Konstruktionen, Springer, 1988.

Module M0603: Nonlinear Structural Analysis

Courses

Title	Type	Hrs/wk	CP
Nonlinear Structural Analysis (L0277)	Lecture	3	4
Nonlinear Structural Analysis (L0279)	Recitation Section (small)	1	2
Module Responsible	Prof. Alexander Düster		
Admission Requirements	None		
Recommended Previous Knowledge	Knowledge of partial differential equations is recommended.		
Educational Objectives	After taking part successfully, students have reached the following learning results		
Professional Competence <i>Knowledge</i> <i>Skills</i> Personal Competence <i>Social Competence</i> <i>Autonomy</i>	Students are able to + give an overview of the different nonlinear phenomena in structural mechanics. + explain the mechanical background of nonlinear phenomena in structural mechanics. + to specify problems of nonlinear structural analysis, to identify them in a given situation and to explain their mathematical and mechanical background. Students are able to + model nonlinear structural problems. + select for a given nonlinear structural problem a suitable computational procedure. + apply finite element procedures for nonlinear structural analysis. + critically verify and judge results of nonlinear finite elements. + to transfer their knowledge of nonlinear solution procedures to new problems. Students are able to + solve problems in heterogeneous groups. + present and discuss their results in front of others. + give and accept professional constructive criticism. Students are able to + assess their knowledge by means of exercises and E-Learning. + acquaint themselves with the necessary knowledge to solve research oriented tasks. + to transform the acquired knowledge to similar problems.		
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56		
Credit points	6		
Course achievement	None		
Examination	Written exam		
Examination duration and scale	120 min		
Assignment for the Following Curricula	Civil Engineering: Specialisation Structural Engineering: Elective Compulsory Civil Engineering: Specialisation Computational Engineering: Compulsory Computational Engineering: Core Qualification: Elective Compulsory International Management and Engineering: Specialisation II. Civil Engineering: Elective Compulsory Mechanical Engineering - Product Development and Production: Core Qualification: Elective Compulsory Materials Science and Engineering: Specialisation Modeling: Elective Compulsory Materials Science: Specialisation Modeling: Elective Compulsory Mechatronics: Technical Complementary Course: Elective Compulsory Mechatronics: Core Qualification: Elective Compulsory Product Development, Materials and Production: Core Qualification: Elective Compulsory Naval Architecture and Ocean Engineering: Core Qualification: Elective Compulsory Naval Architecture and Ocean Engineering: Core Qualification: Elective Compulsory Ship and Offshore Technology: Core Qualification: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Simulation Technology: Elective Compulsory		

Course L0277: Nonlinear Structural Analysis	
Typ	Lecture
Hrs/wk	3
CP	4
Workload in Hours	Independent Study Time 78, Study Time in Lecture 42
Lecturer	Prof. Alexander Düster
Language	EN
Cycle	WiSe
Content	1. Introduction 2. Nonlinear phenomena 3. Mathematical preliminaries 4. Basic equations of continuum mechanics 5. Spatial discretization with finite elements 6. Solution of nonlinear systems of equations 7. Solution of elastoplastic problems 8. Stability problems 9. Contact problems
Literature	[1] Alexander Düster, Nonlinear Structural Analysis, Lecture Notes, Technische Universität Hamburg-Harburg, 2014. [2] Peter Wriggers, Nonlinear Finite Element Methods, Springer 2008. [3] Peter Wriggers, Nichtlineare Finite-Elemente-Methoden, Springer 2001. [4] Javier Bonet and Richard D. Wood, Nonlinear Continuum Mechanics for Finite Element Analysis, Cambridge University Press, 2008.

Course L0279: Nonlinear Structural Analysis	
Typ	Recitation Section (small)
Hrs/wk	1
CP	2
Workload in Hours	Independent Study Time 46, Study Time in Lecture 14
Lecturer	Prof. Alexander Düster
Language	EN
Cycle	WiSe
Content	See interlocking course
Literature	See interlocking course

Module M0658: Innovative CFD Approaches

Courses

Title	Type	Hrs/wk	CP
Application of Innovative CFD Methods in Research and Development (L0239)	Lecture	2	3
Application of Innovative CFD Methods in Research and Development (L1685)	Recitation Section (small)	2	3
Module Responsible	Prof. Thomas Rung		
Admission Requirements	None		
Recommended Previous Knowledge	Students should have sound knowledge of engineering mathematics (series expansions, internal & vector calculus), and be familiar with the foundations of partial/ordinary differential equations. They are expected to be familiar with engineering fluid mechanics. Basic knowledge of numerical analysis or computational fluid dynamics, e.g. acquired in previous CFD courses, is of advantage but not necessary.		
Educational Objectives	After taking part successfully, students have reached the following learning results		
Professional Competence <i>Knowledge</i>	Students will acquire a deeper knowledge of recent trends in computational fluid dynamics (CFD), i.e. finite volume, smoothed particle hydrodynamics and lattice Boltzmann approaches, and can relate recent innovations with present challenges in computational fluid mechanics. They are familiar with the similarities and differences between different Eulerian and Lagrangian discretisation and approximation concepts for investigating on the basis of continuum and kinetic theories. Students have the required knowledge to develop, explain, code and apply numerical models concepts to approximate multiphase and multifield problems with grid and particle based methods, respectively. Students know the fundamentals of simulation based PDE constraint optimisation.		
<i>Skills</i>	The students are able choose and apply appropriate discretisation concepts and flow physics models. They acquire the ability to code computational algorithms dedicated to finite volumes on unstructured grids & particle-based discretisations & structured lattice Boltzmann arrangements, apply these codes for parameter investigations and supplement interfaces to extract simulation data for an engineering analysis. They are able to sophisticatedly judge different solution strategies.		
Personal Competence <i>Social Competence</i>	The students are able to discuss problems, present the results of their own analysis, and jointly develop, implement and report on solution strategies that address given technical reference problems in a team. They to lead team sessions and present solutions to experts.		
<i>Autonomy</i>	The students can independently analyse innovative methods to solving fluid engineering problems. They are able to critically analyse own results as well as external data with regards to the plausibility and reliability. Students are able to structure and perform a simulation-based investigation.		
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56		
Credit points	6		
Course achievement	Compulsory Yes	Bonus 20 %	Form Written elaboration
Examination	Oral exam		
Examination duration and scale	30 min		
Assignment for the Following Curricula	Computational Engineering: Core Qualification: Elective Compulsory Energy Systems: Core Qualification: Elective Compulsory Naval Architecture and Ocean Engineering: Core Qualification: Elective Compulsory Naval Architecture and Ocean Engineering: Core Qualification: Elective Compulsory Ship and Offshore Technology: Core Qualification: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Simulation Technology: Elective Compulsory Process Engineering: Specialisation Process Engineering: Elective Compulsory		

Course L0239: Application of Innovative CFD Methods in Research and Development

Type	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Thomas Rung
Language	DE/EN
Cycle	WiSe
Content	Computational Optimisation, Parallel Computing, Efficient CFD-Procedures for GPU Architectures, Alternative Approximations (Lattice-Boltzmann Methods, Particle Methods), Fluid/Structure-Interaction, Modelling of Hybrid Continua
Literature	Vorlesungsmaterialien /lecture notes

Course L1685: Application of Innovative CFD Methods in Research and Development	
Typ	Recitation Section (small)
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Thomas Rung
Language	DE/EN
Cycle	WiSe
Content	See interlocking course
Literature	See interlocking course

Module M1146: Ship Vibration				
Courses				
Title		Typ	Hrs/wk	CP
Ship Vibration (L1528)		Lecture	2	3
Ship Vibration (L1529)		Recitation Section (small)	2	3
Module Responsible	Dr. Rüdiger Ulrich Franz von Bock und Polach			
Admission Requirements	None			
Recommended Previous Knowledge	Mechanis I - III			
	Structural Analysis of Ships I			
	Fundamentals of Ship Structural Design			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence				
Knowledge				
Skills				
Personal Competence				
Social Competence				
Autonomy	The students are able to communicate and cooperate in a professional environment in the shipbuilding and component supply industry.			
	Students are able to detect vibration-prone components on ships, to model the structure, to select suitable calculation methods and to assess the results			
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56			
Credit points	6			
Course achievement	None			
Examination	Written exam			
Examination duration and scale	3 hours			
Assignment for the Following Curricula	Energy Systems: Specialisation Marine Engineering: Elective Compulsory Naval Architecture and Ocean Engineering: Core Qualification: Compulsory Naval Architecture and Ocean Engineering: Core Qualification: Compulsory Ship and Offshore Technology: Core Qualification: Compulsory Theoretical Mechanical Engineering: Specialisation Maritime Technology: Elective Compulsory			

Course L1528: Ship Vibration	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Dr. Rüdiger Ulrich Franz von Bock und Polach
Language	EN
Cycle	WiSe
Content	1. Introduction; assessment of vibrations 2. Basic equations 3. Beams with discrete / distributed masses 4. Complex beam systems 5. Vibration of plates and Grillages 6. Deformation method / practical hints / measurements 7. Hydrodynamic masses 8. Spectral method 9. Hydrodynamic masses acc. to Lewis 10. Damping 11. Shaft systems 12. Propeller excitation 13. Engines
Literature	Siehe Vorlesungsskript

Course L1529: Ship Vibration	
Typ	Recitation Section (small)
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Dr. Rüdiger Ulrich Franz von Bock und Polach
Language	EN
Cycle	WiSe
Content	1. Introduction; assessment of vibrations 2. Basic equations 3. Beams with discrete / distributed masses 4. Complex beam systems 5. Vibration of plates and Grillages 6. Deformation method / practical hints / measurements 7. Hydrodynamic masses 8. Spectral method 9. Hydrodynamic masses acc. to Lewis 10. Damping 11. Shaft systems 12. Propeller excitation 13. Engines
Literature	Siehe Vorlesungsskript

Module M1169: Ship Design (JM)				
Courses				
Title		Typ	Hrs/wk	CP
Ship Design (JM) (L1577)		Lecture	2	4
Ship Design (JM) (L1578)		Recitation Section (large)	2	2
Module Responsible	Prof. Stefan Krüger			
Admission Requirements	None			
Recommended Previous Knowledge	The student must have participated at the lecture Ship Design (Bachelor). Ship safety			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence	The student shall link the methods and design principles he learnt in the basic lecture to specific design problems of Container, RoRo, RoPax and Passenger Vessels. At the end of the lecture it is expected that the student can carry out ship designs sucessfully. The most imortant design problems, constraints and methods related to the a.m. ship typs are referenced, based on the list of methods developed in Ship Design (BA). The a.m. ship types serve as reference vessels where the application shall point out specific design aspects. The lecture closes with a brief introduction of design principles of dry bulk carriers, paper carriers and ouble ended ferries. The student learns to make technical decisions and to get acceptance for his decisions. Autonomous applicatin of basic technical knowledge to the design process.			
Knowledge				
Skills				
Personal Competence				
Social Competence				
Autonomy				
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56			
Credit points	6			
Course achievement	None			
Examination	Oral exam			
Examination duration and scale	45 min			
Assignment for the Following Curricula	Ship and Offshore Technology: Core Qualification: Compulsory			

Course L1577: Ship Design (JM)	
Typ	Lecture
Hrs/wk	2
CP	4
Workload in Hours	Independent Study Time 92, Study Time in Lecture 28
Lecturer	Prof. Stefan Krüger
Language	DE/EN
Cycle	WiSe
Content	The student shall link the methods and design principles he learnt in the lecture Ship Design I to specific design problems of Container, RoRo, RoPax and Passenger Vessels. At the end of the lecture it is expected that the student can carry out ship designs successfully. The most important design problems, constraints and methods related to the a.m. ship types are referenced, based on the list of methods developed in Ship Design I. The a.m. ship types serve as reference vessels where the application shall point out specific design aspects. The lecture closes with a brief introduction of design principles of dry bulk carriers, paper carriers and double ended ferries.
Literature	Schneekluth: Entwerfen von Schiffen

Course L1578: Ship Design (JM)	
Typ	Recitation Section (large)
Hrs/wk	2
CP	2
Workload in Hours	Independent Study Time 32, Study Time in Lecture 28
Lecturer	Prof. Stefan Krüger
Language	DE/EN
Cycle	WiSe
Content	See interlocking course
Literature	See interlocking course

Module M1176: Seakeeping of Ships and Laboratory on Naval Architecture				
Courses				
Title		Type	Hrs/wk	CP
Laboratory on Naval Architecture (L0241)		Practical Course	2	2
Seakeeping of Ships (L1594)		Lecture	2	3
Seakeeping of Ships (L1619)		Recitation Section (small)	2	1
Module Responsible	Prof. Moustafa Abdel-Maksoud			
Admission Requirements	None			
Recommended Previous Knowledge	Basic knowledge of ship dynamics as well as stochastic and statistics			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence <i>Knowledge</i>	• Understand present research questions in the field of ship motion in waves • Explain the present state of the art for the topics considered • Apply given methodology to approach given problems of seakeeping behavior • Evaluate the limits of the present methods • Identify possibilities to extend present methods • Evaluate the feasibility of further developments			
	<i>Skills</i>	Students are able to • select and apply suitable computing and simulation methods to determine the dynamic loads on ships and floating bodies • model the behavior of ships and floating bodies under different sea conditions by using simplified methods • evaluate critically the investigation results of experimental or numerical studies		
	Personal Competence <i>Social Competence</i>	Students are able to • solve problems in heterogeneous groups and to document the corresponding results • share new knowledge with group members		
	<i>Autonomy</i>	Students are able to • assess their knowledge by means of exercises • think system-oriented • decompose complex systems		
Workload in Hours	Independent Study Time 96, Study Time in Lecture 84			
Credit points	6			
Course achievement	Compulsory	Bonus	Form	Description
	Yes	20 %	Exercises	
Examination	Written exam			
Examination duration and scale	180 min			
Assignment for the Following Curricula	Naval Architecture and Ocean Engineering: Core Qualification: Compulsory Naval Architecture and Ocean Engineering: Core Qualification: Compulsory Ship and Offshore Technology: Core Qualification: Elective Compulsory			

Course L0241: Laboratory on Naval Architecture	
Typ	Practical Course
Hrs/wk	2
CP	2
Workload in Hours	Independent Study Time 32, Study Time in Lecture 28
Lecturer	Prof. Thomas Rung, Hauke Herrnring
Language	DE/EN
Cycle	SoSe
Content	The lab is structured into 5 team-based experiments 1. Resistance test Towing test to investigate a model hull resistance 2. Propulsion test Propulsion test for a self propelled hull. Determination of thrust deduction, wake fraction and propulsion efficiency. 3. Seakeeping test Investigation of the seakeeping behaviour 4. Open water and cavitation test Compilation of an open water diagram and cavitation experiments 5. Application of strain measurement techniques Theoretical instructions will also involve foundations of similarity analysis
Literature	Vorlesungsmanuskript Lecture Notes

Course L1594: Seakeeping of Ships	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Moustafa Abdel-Maksoud
Language	DE/EN
Cycle	WiSe
Content	1. Numerical methods for the determination of section forces 2. Steep waves (Stokes-Theory) 3. 3d-potential flow methods 4. Time domain simulation of ship motions 5. Capsizing 6. Slamming
Literature	• Söding, H., Schiffe im Seegang I, Vorlesungsmanuskript, Institut für Fluidodynamik und Schiffstheorie, TUHH, Hamburg, 1992 • Jensen, G., Söding, H. S., Schiffe im Seegang II, Vorlesungsmanuskript, Institut für Fluidodynamik und Schiffstheorie, TUHH, Hamburg, 2005 • Bertram, V., Practical Ship Design Hydrodynamics, Butterworth-Heinemann, Linacre House, Jordan Hill, Oxford, United Kingdom, 2000 • Lloyd, A., Ship Behaviour in Rough Weather, Gosport, Chichester, Sussex, United Kingdom, 1998 • Jensen, J. J., Load and Global Response of Ships, Elsevier Science, Oxford, United Kingdom, 2001

Course L1619: Seakeeping of Ships	
Typ	Recitation Section (small)
Hrs/wk	2
CP	1
Workload in Hours	Independent Study Time 2, Study Time in Lecture 28
Lecturer	Prof. Moustafa Abdel-Maksoud
Language	DE/EN
Cycle	WiSe
Content	See interlocking course
Literature	See interlocking course

Course L1598: Shallow Water Ship Hydrodynamics	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Moustafa Abdel-Maksoud, Dr. Norbert Stuntz
Language	DE/EN
Cycle	WiSe
Content	• Special Aspects of Shallow Water Hydrodynamics, Vertical and Horizontal Constraints, Irregularities in Channel Bed • Fundamental Equations of Shallow Water Hydrodynamics • Approximation of Shallow Water Waves, Boussinesq's Approximation • Ship Waves in Deep Water and under critical, non-critical and supercritical Velocities • Solitary Wves, Critical Speed Range, Extinction of Waves • Aspects of Ship motions in Canals with limited water depth
Literature	• PNA (1988): Principle of Naval Architecture, Vol. II, ISBN 0-939773-01-5 • Schneekluth (1988): Hydromechanik zum Schiffsentwurf • Jiang, T. (2001): Ship Waves in Shallow Water, Fortschritt-Berichte VDI, Series 12, No 466, ISBN 3-18-346612-0

Module M1232: Arctic Technology				
Courses				
Title	Typ		Hrs/wk	CP
Ice Engineering (L1607)	Lecture		2	2
Ice Engineering (L1615)	Recitation Section (small)		1	2
Ship structural design for arctic conditions (L1575)	Project-/problem-based Learning		2	2
Module Responsible	Prof. Sören Ehlers			
Admission Requirements	None			
Recommended Previous Knowledge	none			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence	<i>Knowledge</i> The challenges and requirements due to ice can be explained. Ice loads can be explained and ice strengthening can be understood. <i>Skills</i> The challenges and requirements due to ice can be assessed and the accuracy of these assessment can be evaluated. Calculation models to assess ice loads can be used and a structure can be designed accordingly. <i>Personal Competence</i> <i>Social Competence</i> Students are capable to present their structural design and discuss their decisions constructively in a group. <i>Autonomy</i> Independent and individual assignment tasks can be carried out and presented whereby the capabilities to both, present and defend, the skills and findings will be achieved.			
Workload in Hours	Independent Study Time 110, Study Time in Lecture 70			
Credit points	6			
Course achievement	None			
Examination	Oral exam			
Examination duration and scale	30 min			
Assignment for the Following Curricula	Naval Architecture and Ocean Engineering: Core Qualification: Elective Compulsory Naval Architecture and Ocean Engineering: Core Qualification: Elective Compulsory Ship and Offshore Technology: Core Qualification: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Maritime Technology: Elective Compulsory			

Course L1607: Ice Engineering	
Typ	Lecture
Hrs/wk	2
CP	2
Workload in Hours	Independent Study Time 32, Study Time in Lecture 28
Lecturer	Dr. Walter Kuehnlein
Language	DE/EN
Cycle	WiSe
Content	- Ice, Ice Properties, Ice Failure Modes and Challenges and Requirements due to Ice - Introduction, what is/means ice engineering - Description of different kinds of ice, main ice properties and different ice failure modes - Why is ice so different compared to open water - Presentation of design challenges and requirements for structures and systems in ice covered waters - Ice Load Determination and Ice Model Testing - Overview of different empirical equations for simple determination of ice loads - Discussion and interpretation of the different equations and results - Introduction to ice model tests - What are the requirements for ice model tests, what parameters have to be scaled - What can be simulated and how to use the results of such ice model tests - Computational Modelling of Ice-Structure Interaction Processes - Dynamic fracture and continuum mechanics for modelling ice-structure interaction processes - Alternative numerical crack propagation modelling methods. Examples of cohesive element models for real life structures. - Discussion of contribution of ice properties, hydrodynamics and rubble. - Ice Design Philosophies and Perspectives - What has to be considered when designing structures or systems for ice covered waters - What are the main differences compared to open water design - Ice Management - What are the main ice design philosophies and why is an integrated concept so important for ice Learning Objectives The course will provide an introduction into ice engineering. Different kinds of ice and their different failure modes including numerical methods for ice load simulations are presented. Main design issues including design philosophies for structures and systems for ice covered waters are introduced. The course shall enable the attendees to understand the fundamental challenges due to ice covered waters and help them to understand ice engineering reports and presentations.
Literature	- Proceedings OMAE - Proceedings POAC - Proceedings ATC

Course L1615: Ice Engineering	
Typ	Recitation Section (small)
Hrs/wk	1
CP	2
Workload in Hours	Independent Study Time 46, Study Time in Lecture 14
Lecturer	Dr. Walter Kuehnlein
Language	DE/EN
Cycle	WiSe
Content	See interlocking course
Literature	See interlocking course

Course L1575: Ship structural design for arctic conditions	
Typ	Project-/problem-based Learning
Hrs/wk	2
CP	2
Workload in Hours	Independent Study Time 32, Study Time in Lecture 28
Lecturer	Dr. Rüdiger Ulrich Franz von Bock und Polach, Dr. Rüdiger Ulrich Franz von Bock und Polach
Language	DE/EN
Cycle	WiSe
Content	The structural design under ice loads will be carried out for an individual case
Literature	FSICR, IACS PC and assorted publications

Module M1240: Fatigue Strength of Ships and Offshore Structures

Courses

Title	Typ	Hrs/wk	CP
Fatigue Strength of Ships and Offshore Structures (L1521)	Lecture	2	3
Fatigue Strength of Ships and Offshore Structures (L1522)	Recitation Section (small)	2	3
Module Responsible	Prof. Sören Ehlers		
Admission Requirements	None		
Recommended Previous Knowledge	Structural analysis of ships and/or offshore structures and fundamental knowledge in mechanics and mechanics of materials		
Educational Objectives	After taking part successfully, students have reached the following learning results		
Professional Competence	<i>Knowledge</i> Students are able to - describe fatigue loads and stresses, as well as - describe structural behaviour under cyclic loads. <i>Skills</i> Students are able to calculate life prediction based on the S-N approach as well as life prediction based on the crack propagation. <i>Personal Competence</i> <i>Social Competence</i> The students are able to communicate and cooperate in a professional environment in the shipbuilding and component supply industry. <i>Autonomy</i> The widespread scope of gained knowledge enables the students to handle situations in their future profession independently and confidently.		
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56		
Credit points	6		
Course achievement	None		
Examination	Oral exam		
Examination duration and scale	30 min		
Assignment for the Following Curricula	Naval Architecture and Ocean Engineering: Core Qualification: Elective Compulsory Naval Architecture and Ocean Engineering: Core Qualification: Elective Compulsory Ship and Offshore Technology: Core Qualification: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Maritime Technology: Elective Compulsory		

Course L1521: Fatigue Strength of Ships and Offshore Structures

Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Dr. Moritz Braun
Language	EN
Cycle	WiSe
Content	1.) Introduction 2.) Fatigue loads and stresses 3.) Structural behaviour under cyclic loads - Structural behaviour under constant amplitude loading - Influence factors on fatigue strength - Material behaviour under constant amplitude loading - Special aspects of welded joints - Structural behaviour under variable amplitude loading 4.) Life prediction based on the S-N approach - Damage accumulation hypotheses - nominal stress approach - structural stress approach - notch stress approach - notch strain approach - numerical analyses 5.) Life prediction based on the crack propagation - basic relationships in fracture mechanics - description of crack propagation - numerical analysis - safety against unstable fracture
Literature	Siehe Vorlesungsskript

Course L1522: Fatigue Strength of Ships and Offshore Structures	
Typ	Recitation Section (small)
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Dr. Moritz Braun
Language	EN
Cycle	WiSe
Content	See interlocking course
Literature	See interlocking course

Thesis

Module M-002: Master Thesis

Courses

Title	Typ	Hrs/wk	CP
Module Responsible	Professoren der TUHH		
Admission Requirements	According to General Regulations §21 (1): At least 60 credit points have to be achieved in study programme. The examinations board decides on exceptions.		
Recommended Previous Knowledge			
Educational Objectives	After taking part successfully, students have reached the following learning results		
Professional Competence <i>Knowledge</i>	- The students can use specialized knowledge (facts, theories, and methods) of their subject competently on specialized issues. - The students can explain in depth the relevant approaches and terminologies in one or more areas of their subject, describing current developments and taking up a critical position on them. - The students can place a research task in their subject area in its context and describe and critically assess the state of research.		
Skills	The students are able: - To select, apply and, if necessary, develop further methods that are suitable for solving the specialized problem in question. - To apply knowledge they have acquired and methods they have learnt in the course of their studies to complex and/or incompletely defined problems in a solution-oriented way. - To develop new scientific findings in their subject area and subject them to a critical assessment.		
Personal Competence <i>Social Competence</i>	Students can - Both in writing and orally outline a scientific issue for an expert audience accurately, understandably and in a structured way. - Deal with issues competently in an expert discussion and answer them in a manner that is appropriate to the addressees while upholding their own assessments and viewpoints convincingly.		
<i>Autonomy</i>	Students are able: - To structure a project of their own in work packages and to work them off accordingly. - To work their way in depth into a largely unknown subject and to access the information required for them to do so. - To apply the techniques of scientific work comprehensively in research of their own.		
Workload in Hours	Independent Study Time 900, Study Time in Lecture 0		
Credit points	30		
Course achievement	None		
Examination	Thesis		
Examination duration and scale	According to General Regulations		
Assignment for the Following Curricula	Civil Engineering: Thesis: Compulsory Bioprocess Engineering: Thesis: Compulsory Chemical and Bioprocess Engineering: Thesis: Compulsory Chemical and Bioprocess Engineering: Thesis: Compulsory Computational Engineering: Thesis: Compulsory Computer Science: Thesis: Compulsory Data Science: Thesis: Compulsory Electrical Engineering and Information Technology: Thesis: Compulsory Electrical Engineering: Thesis: Compulsory Energy Systems: Thesis: Compulsory Environmental Engineering: Thesis: Compulsory Aircraft Systems Engineering: Thesis: Compulsory Global Innovation Management: Thesis: Compulsory Computer Science in Engineering: Thesis: Compulsory Information and Communication Systems: Thesis: Compulsory Interdisciplinary Mathematics: Thesis: Compulsory International Production Management: Thesis: Compulsory International Management and Engineering: Thesis: Compulsory		

Joint European Master in Environmental Studies - Cities and Sustainability: Thesis: Compulsory
Logistics, Infrastructure and Mobility: Thesis: Compulsory
Aeronautics: Thesis: Compulsory
Mechanical Engineering - Product Development and Production: Thesis: Compulsory
Materials Science and Engineering: Thesis: Compulsory
Materials Science: Thesis: Compulsory
Mechanical Engineering and Management: Thesis: Compulsory
Mechatronics: Thesis: Compulsory
Biomedical Engineering: Thesis: Compulsory
Microelectronics and Microsystems: Thesis: Compulsory
Product Development, Materials and Production: Thesis: Compulsory
Renewable Energies: Thesis: Compulsory
Naval Architecture and Ocean Engineering: Thesis: Compulsory
Naval Architecture and Ocean Engineering: Thesis: Compulsory
Ship and Offshore Technology: Thesis: Compulsory
Theoretical Mechanical Engineering: Thesis: Compulsory
Process Engineering: Thesis: Compulsory
Water and Environmental Engineering: Thesis: Compulsory
Certification in Engineering & Advisory in Aviation: Thesis: Compulsory